Supplementary materials for:

"Antarctic Sanctuary: Fishing Effort Responses to an International MPA in the Southern Ocean"

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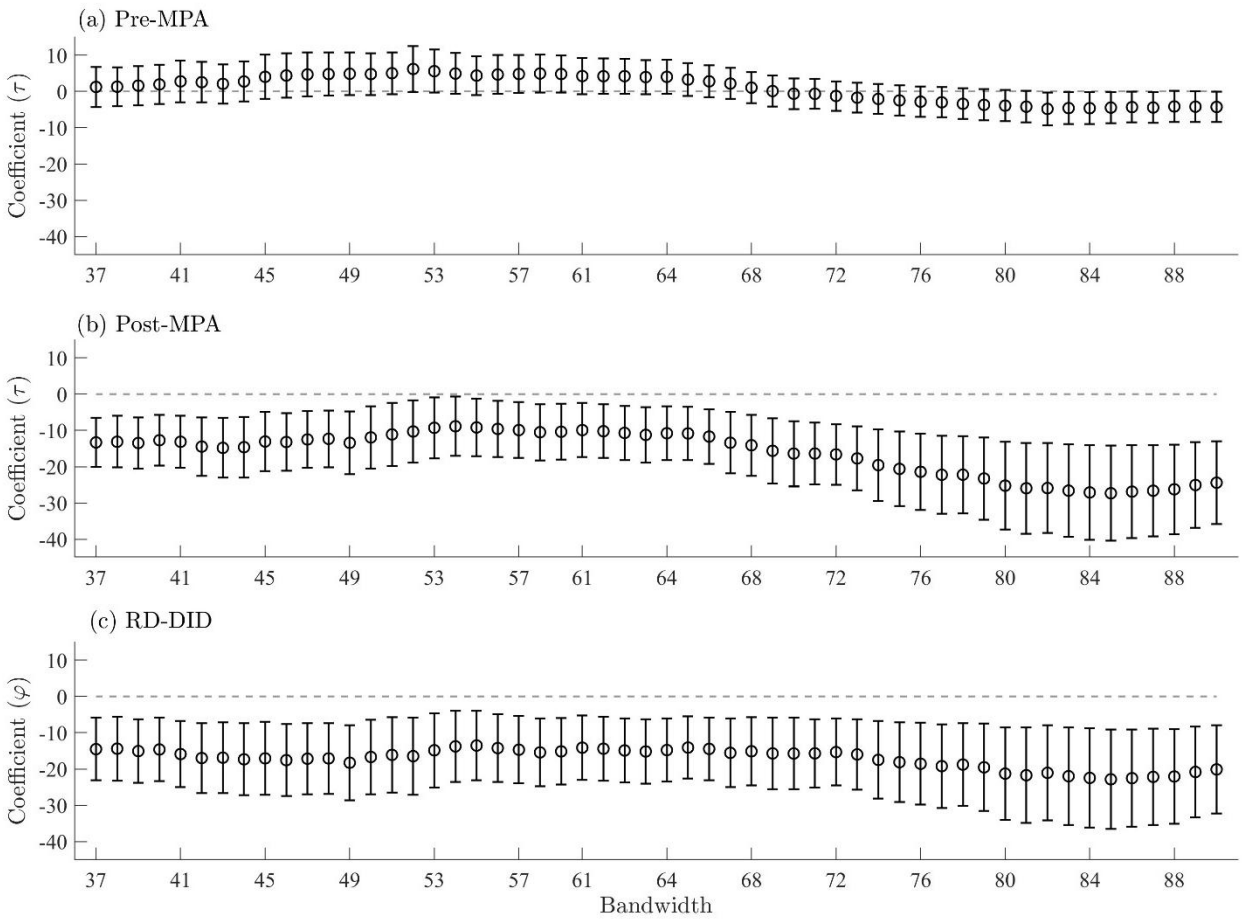
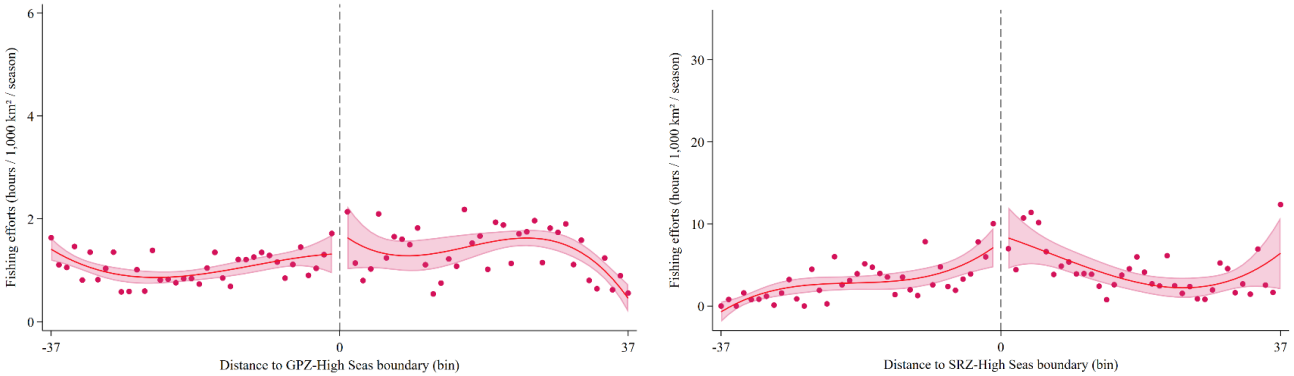
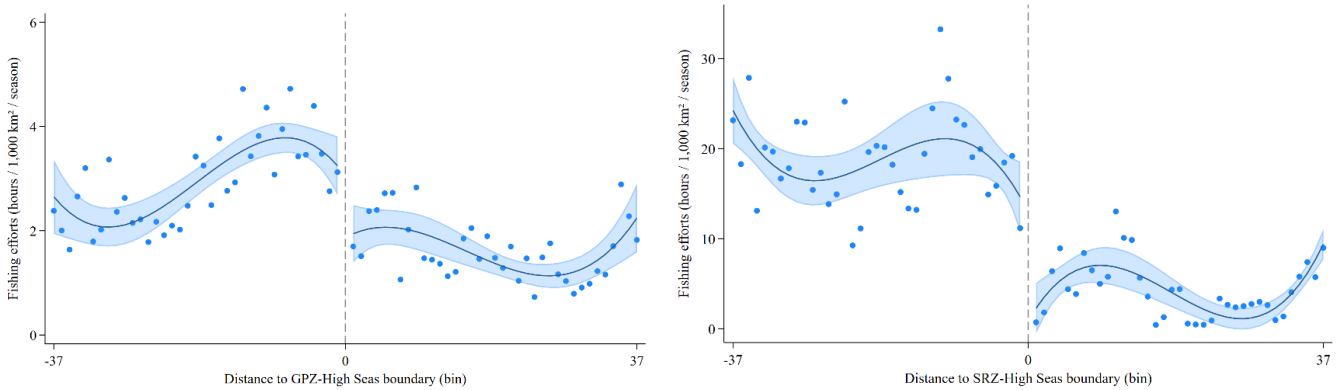


Supplementary Figure 1. Discontinuity in total fishing hours per season per thousand km^2 at the MPA-High Seas boundary during the pre-MPA period (a) and the post-MPA period (b), and the results for the RD-DID regression (c), for varying bandwidths from 37 bins to 90 bins. Estimates of the coefficient τ in Equation (2) for panels (a) and (b), and the coefficient φ in Equation (3) for panel (c), are reported with their 95 confidence intervals.

(a) Pre-MPA



(b) Post-MPA



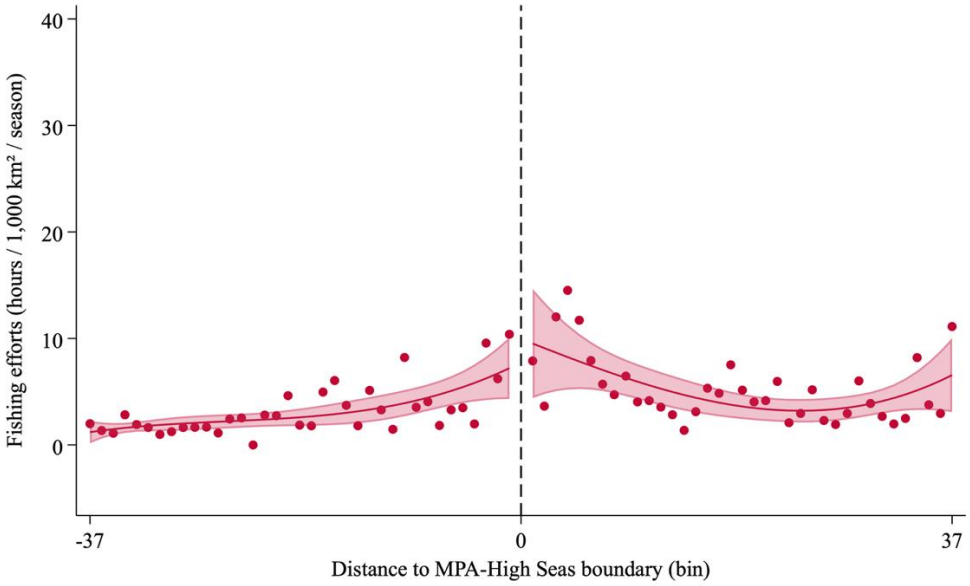
Supplementary Figure 2. Total fishing hours per season per thousand km² within 37 bins of the GPZ and SRZ boundaries during the pre- and post-MPA periods. Lines are third-order polynomial fits in distance to the boundary, and shaded areas are 95% confidence intervals estimated using Newey-West standard errors that account for heteroskedasticity and autocorrelation.

Supplementary Table 1. RD-DID results for the GPZ and SRZ boundaries

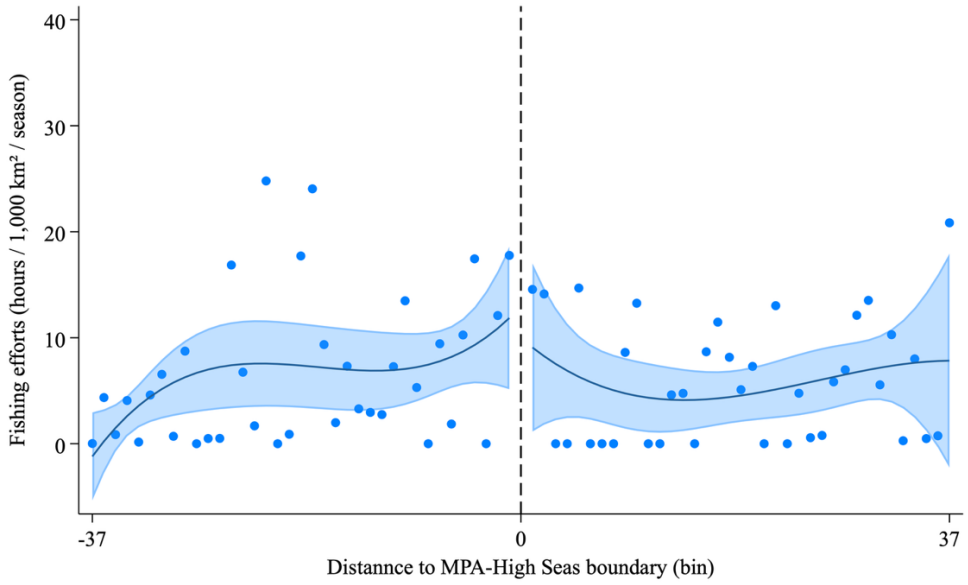
	GPZ		SRZ	
	(1)	(2)	(3)	(4)
Treatment × Post-MPA	-2.464***	-1.657**	-13.739***	-12.921***
($D_{b,p} \times T_p$)	(0.418)	(0.678)	(2.930)	(4.247)
Post-MPA (T_p)	2.935***	1.857***	12.964***	5.183*
	(0.286)	(0.436)	(2.222)	(3.089)
Treatment ($D_{b,p}$)	0.401**	0.421	0.474	0.801
	(0.207)	(0.444)	(1.532)	(2.716)
Polynomial	1	3	1	3
Observations	148	148	148	148

Notes: This table reports the estimation results for the RD-DID regression of Equation (3) for the GPZ-High Seas boundary (columns 1 and 2) and the SRZ-High Seas boundary (columns 3 and 4). The dependent variable is the total fishing hours per season per thousand km² within bin *b*. Each bin has a size of 1.11 km, within a bandwidth of 37 bins. The regressions control for a linear trend and third-order polynomial in distance to the boundary. Newey-West standard errors that account for heteroskedasticity and autocorrelation are reported in parentheses. Significance levels at 10%, 5%, and 1% are denoted by *, **, and ***, respectively.

(a) Pre-announcement of MPA (1 January 2012 to 27 October 2016)



(b) Post-announcement, pre-implementation of MPA (28 October 2016 to 30 November 2017)

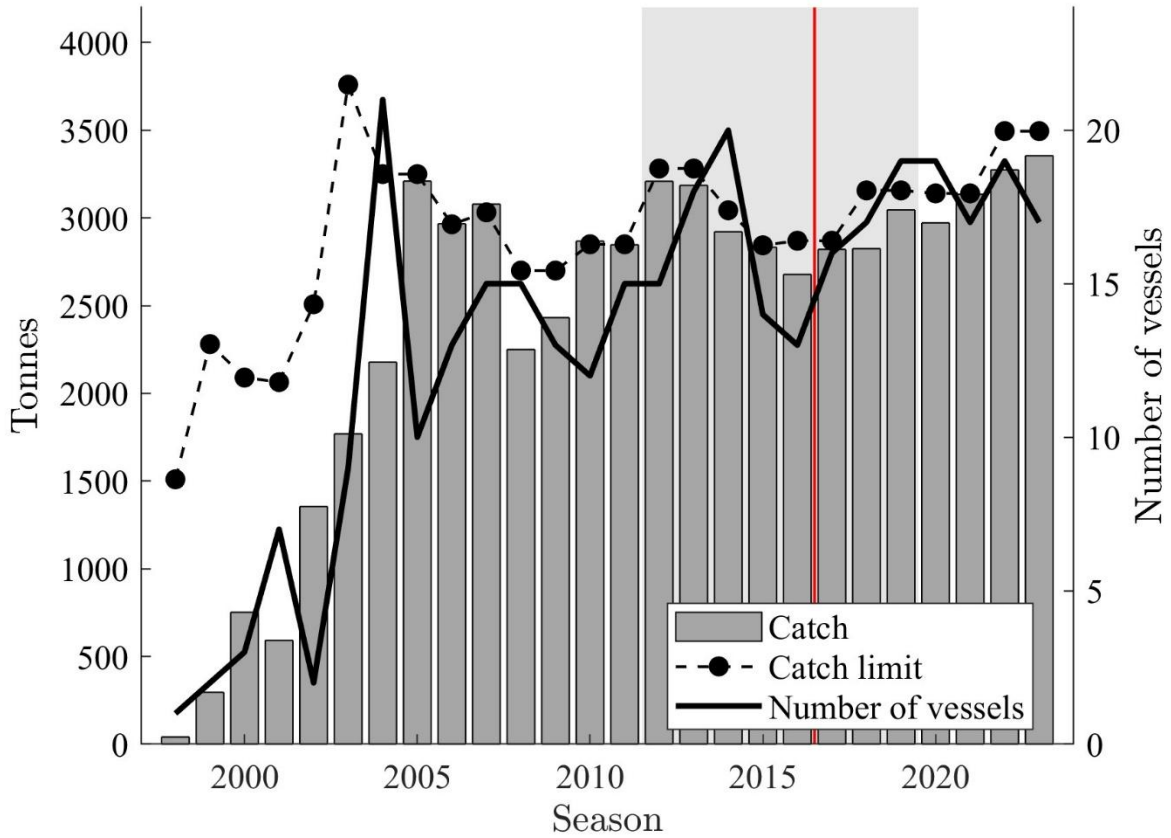


Supplementary Figure 3. Total fishing hours per season per thousand km² within 37 bins of the MPA-High Seas boundary during the pre-announcement period (a) and the post-announcement-to-implementation period (b). Each bin has a size of 1.11 km, within the bandwidth of 37 bins. Lines are third-order polynomial fits in distance to the boundary, and shaded areas are 95% confidence intervals estimated using Newey-West standard errors that account for heteroskedasticity and autocorrelation.

Supplementary Table 2. Discontinuity in the ocean productivity index and ocean depth

	Ocean productivity index			Ocean depth
	(1) Pre-MPA	(2) Post-MPA	(3) RD-DID	(4)
Treatment × Post-MPA ($D_{b,p} \times T_p$)			-1.288	
			(21.371)	
Post-MPA (T_p)			10.16	
			(16.086)	
Treatment ($D_{b,p}$)	-3.028	-3.178	-1.74	-21.512
	(17.175)	(14.799)	(12.727)	(16.101)
Polynomial	3	3	3	3
Observations	74	74	148	74

Notes: This table reports the estimation results when the dependent variable is either the mean ocean productivity index per day per km² or mean ocean depth in meters within bin *b*, as reported in the column headings. Each bin has a size of 1.11 km, within a bandwidth of 37 km. The regressions control for a third-order polynomial in distance to the boundary. Newey-West standard errors that account for heteroskedasticity and autocorrelation are reported in parentheses. Significance levels at 10%, 5%, and 1% are denoted by *, **, and ***, respectively.



Supplementary Figure 4. Catch and catch limit for toothfish and number of vessels in the Ross Sea region (subareas 88.1 and 882A-B). The time at which the Ross Sea MPA was implemented is indicated by a red vertical line. The period during which the GFW fishing effort data were used in the analysis is shown by the shaded area. Data sources: CCAMLR Statistical Bulletin and CCAMLR (2024).